

# ENGR 213 · Applied Ordinary Differential Equations

## Master Step-by-Step Solutions & Method Expansions Manual

 **Concordia University** (BCEE Department)

 **Standard:** Universal Step-by-Step Expansion Format (exact\_ode\_step\_by\_step.pdf standard)

 **Scope:** Lectures 1–5 · Tutorial 1 · Quiz 1 & Midterm 1 Coverage

 **The Step-by-Step Expansion Philosophy:** Every single problem is structured into numbered steps with explicit theory stated before computation, zero skipped algebra, and a standardized **"Pattern to Remember"** algorithm box at the end.

**Problem 1 · Exact Differential Equation (Potential Function Method)****Problem:** Solve  $2xy \, dx + (x^2 - 1) \, dy = 0$ **Step 1: Identify  $M(x,y)$  and  $N(x,y)$** The standard form is  $M(x, y) \, dx + N(x, y) \, dy = 0$ . Comparing with the given equation, we get:

$$M(x, y) = 2xy \quad \text{and} \quad N(x, y) = x^2 - 1$$

**Step 2: Check whether the equation is exact**For an equation to be exact, the partial derivatives must satisfy  $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$ .

$$\frac{\partial M}{\partial y} = \frac{\partial}{\partial y}[2xy] = 2x, \quad \frac{\partial N}{\partial x} = \frac{\partial}{\partial x}[x^2 - 1] = 2x$$

Because  $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} = 2x$ , the equation is exact.**Step 3: Find the potential function  $F(x,y)$** For an exact equation, there exists  $F(x, y)$  such that  $\frac{\partial F}{\partial x} = M$  and  $\frac{\partial F}{\partial y} = N$ . Integrate with respect to  $x$  (treating  $y$  as constant):

$$F(x, y) = \int 2xy \, dx = y \int 2x \, dx = x^2 y + g(y)$$

The extra term  $g(y)$  is included because the constant of integration may depend on  $y$ .**Step 4: Differentiate  $F$  with respect to  $y$** 

$$\frac{\partial F}{\partial y} = x^2 + g'(y)$$

Equating to  $N(x, y) = x^2 - 1$ :

$$x^2 + g'(y) = x^2 - 1 \implies g'(y) = -1$$

**Step 5: Find  $g(y)$** 

$$g(y) = \int (-1) \, dy = -y$$

**Step 6: Write the implicit and explicit solution**Substitute  $g(y) = -y$  into  $F(x, y) = C$ :

$$x^2 y - y = C \implies y(x^2 - 1) = C \implies \mathbf{y(x) = \frac{C}{x^2 - 1}}$$

**★ Exact ODE Pattern to Remember:**

1. Identify  $M$  and  $N$  from  $M \, dx + N \, dy = 0$ .
2. Check  $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$ .
3. Integrate  $M$  with respect to  $x$ :  $F(x, y) = \int M \, dx + g(y)$ .
4. Differentiate  $F$  with respect to  $y$  and equate to  $N$  to isolate  $g'(y)$ .
5. Integrate  $g'(y)$  to find  $g(y)$ , and write the solution  $F(x, y) = C$ .

## Problem 2 · Non-Exact ODE (Integrating Factor $\mu(x)$ )

Lecture 4 Slide 10

**Problem:** Solve  $(2x^2 + y) dx + (x^2y - x) dy = 0$

### Step 1: Identify M and N

$$M = 2x^2 + y, \quad N = x^2y - x$$

### Step 2: Check for exactness

$$\frac{\partial M}{\partial y} = 1, \quad \frac{\partial N}{\partial x} = 2xy - 1$$

Since  $1 \neq 2xy - 1$ , the equation is not exact.

### Step 3: Find the integrating factor

$$\frac{M_y - N_x}{N} = \frac{1 - (2xy - 1)}{x(xy - 1)} = \frac{-2(xy - 1)}{x(xy - 1)} = -\frac{2}{x}$$

$$\mu(x) = e^{\int -2/x dx} = e^{-2 \ln |x|} = \frac{1}{x^2}$$

### Step 4: Multiply by $\mu(x)$ and verify exactness

$$\left(2 + \frac{y}{x^2}\right) dx + \left(y - \frac{1}{x}\right) dy = 0 \implies \tilde{M}_y = \frac{1}{x^2} = \tilde{N}_x \quad (\text{Exact!})$$

### Step 5: Integrate $\tilde{M}$ to find $F(x,y)$

$$F(x, y) = \int \left(2 + \frac{y}{x^2}\right) dx = 2x - \frac{y}{x} + g(y)$$

### Step 6: Match $\frac{\partial F}{\partial y}$ to $\tilde{N}$

$$-\frac{1}{x} + g'(y) = y - \frac{1}{x} \implies g'(y) = y \implies g(y) = \frac{1}{2}y^2$$

### Step 7: Final implicit solution

$$2x - \frac{y}{x} + \frac{1}{2}y^2 = C$$

#### ★ Integrating Factor Pattern to Remember:

1. If  $M_y \neq N_x$ , compute  $\frac{M_y - N_x}{N}$ . If it depends only on  $x$ ,  $\mu(x) = e^{\int \frac{M_y - N_x}{N} dx}$ .
2. Multiply the entire ODE by  $\mu(x)$  to obtain exact form  $\tilde{M} dx + \tilde{N} dy = 0$ .
3. Integrate  $\tilde{M}$  with respect to  $x$  and solve for  $g(y)$  by equating  $F_y = \tilde{N}$ .

### Problem 3 · Verification of a Two-Parameter Solution Family

Tutorial 1 · Sec 1.1 #25

**Problem:** Verify that  $y = c_1 e^{2x} + c_2 x e^{2x}$  satisfies  $y'' - 4y' + 4y = 0$  on  $(-\infty, \infty)$ .

#### Step 1: Compute first derivative $y'(x)$

Apply chain rule on  $c_1 e^{2x}$  and product rule on  $c_2 x e^{2x}$ :

$$y'(x) = 2c_1 e^{2x} + c_2 (e^{2x} + 2x e^{2x}) = (2c_1 + c_2) e^{2x} + 2c_2 x e^{2x}$$

#### Step 2: Compute second derivative $y''(x)$

$$y''(x) = 2(2c_1 + c_2) e^{2x} + 2c_2 (e^{2x} + 2x e^{2x}) = (4c_1 + 4c_2) e^{2x} + 4c_2 x e^{2x}$$

#### Step 3: Substitute into LHS and group like terms

$$\begin{aligned} \text{LHS} &= y'' - 4y' + 4y \\ &= [(4c_1 + 4c_2) e^{2x} + 4c_2 x e^{2x}] - 4[(2c_1 + c_2) e^{2x} + 2c_2 x e^{2x}] + 4[c_1 e^{2x} + c_2 x e^{2x}] \\ &= e^{2x} (4c_1 + 4c_2 - 8c_1 - 4c_2 + 4c_1) + x e^{2x} (4c_2 - 8c_2 + 4c_2) \\ &= e^{2x} (0) + x e^{2x} (0) = 0 = \text{RHS} \quad \blacksquare \end{aligned}$$

#### ✦ Verification Pattern to Remember:

1. Differentiate  $y(x)$  matching the order of the ODE (use product rule on  $x e^{kx}$ ).
2. Substitute derivatives into the LHS only.
3. Group coefficients of each independent functional term and demonstrate algebraic cancellation to 0.

### Problem 4 · First-Order Non-Linear IVP & Maximal Interval of Definition

Tutorial 1 · Sec 1.2 #1

**Problem:** Given  $y = \frac{1}{1+c_1e^{-x}}$  satisfies  $y' = y - y^2$ , solve with  $y(0) = -\frac{1}{3}$  and find interval  $I$ .

**Step 1: Apply initial condition to solve for  $c_1$**

$$y(0) = -\frac{1}{3} \implies -\frac{1}{3} = \frac{1}{1+c_1e^0} = \frac{1}{1+c_1} \implies 1+c_1 = -3 \implies c_1 = -4$$

Particular solution:  $y(x) = \frac{1}{1-4e^{-x}}$

**Step 2: Find singularities**

Set denominator equal to zero:

$$1 - 4e^{-x} = 0 \implies 4e^{-x} = 1 \implies e^x = 4 \implies x = \ln 4 \approx 1.386$$

**Step 3: Select the connected interval containing  $x_0 = 0$**

Comparing  $x_0 = 0$  with  $x = \ln 4$ , we see  $0 < \ln 4$ . The connected interval containing  $x_0$  is:

$$I = (-\infty, \ln 4)$$

★ **IVP Interval Pattern to Remember:**

1. Solve for  $c$  using  $(x_0, y_0)$ .
2. Locate vertical asymptotes where denominators = 0.
3. Select the single connected interval containing  $x_0$ . Never write unions ( $\cup$ )!

### Problem 5 · Autonomous DE & Phase Portrait Stability

Tutorial 1 · Sec 2.1 #21

**Problem:** Find critical points, phase line, and stability for  $\frac{dy}{dx} = y^2 - 3y$ .

**Step 1: Find critical points**

Set  $f(y) = y(y - 3) = 0 \implies y = 0$  and  $y = 3$ .

**Step 2: Test interval signs**

- $y > 3$ :  $f(4) = +4 > 0 \implies$  Arrow points UP ( $\uparrow$ ).
- $0 < y < 3$ :  $f(1) = -2 < 0 \implies$  Arrow points DOWN ( $\downarrow$ ).
- $y < 0$ :  $f(-1) = +4 > 0 \implies$  Arrow points UP ( $\uparrow$ ).

**Step 3: Classify stability**

- $y = 3$  is **UNSTABLE (Repeller)** (arrows diverge away:  $\downarrow \bullet \uparrow$ ).
- $y = 0$  is **ASYMPTOTICALLY STABLE (Attractor)** (arrows converge:  $\uparrow \bullet \downarrow$ ).

★ **Autonomous Stability Pattern to Remember:**

1. Set  $f(y) = 0$  and factor to find critical points.
2. Test values between critical points:  $f(y) > 0 \implies \uparrow$ ,  $f(y) < 0 \implies \downarrow$ .
3. Converging arrows  $\implies$  Stable (Attractor); Diverging arrows  $\implies$  Unstable (Repeller).
4. Derivative shortcut:  $f'(c) < 0 \implies$  Stable,  $f'(c) > 0 \implies$  Unstable.

## Problem 6 · Separable ODE & Recovering Singular Lost Solutions

Lecture 3 Slide 6

**Problem:** Solve  $\frac{dy}{dx} = y^2 - 4$  and state any singular solutions.

**Step 1: Check constant equilibrium solutions**

$$y^2 - 4 = 0 \implies (y - 2)(y + 2) = 0 \implies y = 2 \quad \text{and} \quad y = -2$$

**Step 2: Separate variables & partial fractions**

$$\frac{dy}{(y - 2)(y + 2)} = dx \implies \frac{1}{4} \left( \frac{1}{y - 2} - \frac{1}{y + 2} \right) dy = dx$$

**Step 3: Integrate and isolate  $y(x)$**

$$\ln \left| \frac{y - 2}{y + 2} \right| = 4x + C_1 \implies \frac{y - 2}{y + 2} = ce^{4x} \implies y(x) = \frac{2(1 + ce^{4x})}{1 - ce^{4x}}$$

**Step 4: Test for singular lost solutions**

Setting  $c = 0 \implies y = 2$  (included in family). Setting  $y = -2 \implies 1 = -1$  (impossible for any finite  $c$ ). Therefore,  $y \equiv -2$  is a SINGULAR (LOST) SOLUTION.

### ★ Separable ODE Pattern to Remember:

1. Always find  $h(y) = 0$  equilibria *before* dividing.
2. Separate  $\frac{dy}{h(y)} = g(x)dx$  and integrate.
3. Solve explicitly for  $y(x)$ .
4. Check if every equilibrium can be generated by a finite constant  $c$ . If not, state it as a singular solution.

## Problem 7 · First-Order Linear ODE (Integrating Factor)

Lecture 3 Slide 12

**Problem:** Solve the IVP  $x \frac{dy}{dx} + 2y = 4x^2$  with  $y(1) = 2$ .

**Step 1: Put in canonical standard form (divide by  $x$ )**

$$\frac{dy}{dx} + \frac{2}{x}y = 4x \implies P(x) = \frac{2}{x}, \quad Q(x) = 4x$$

**Step 2: Calculate integrating factor  $\mu(x)$**

$$\mu(x) = e^{\int \frac{2}{x} dx} = e^{2 \ln |x|} = e^{\ln(x^2)} = x^2$$

**Step 3: Multiply and integrate**

$$\frac{d}{dx}[x^2 y] = 4x^3 \implies x^2 y = x^4 + C \implies y(x) = x^2 + \frac{C}{x^2}$$

**Step 4: Apply initial condition  $y(1) = 2$**

$$2 = 1^2 + \frac{C}{1^2} \implies C = 1 \implies y(x) = x^2 + \frac{1}{x^2}, \quad I = (0, \infty)$$

### ★ Linear ODE Pattern to Remember:

1. Divide by leading coefficient to get  $\frac{dy}{dx} + P(x)y = Q(x)$ .
2. Compute  $\mu(x) = e^{\int P dx}$ .
3. Integrate product:  $\mu y = \int \mu Q dx + C$ .
4. Divide  $C$  by  $\mu(x)$  to get general solution.

**Problem 8 · Homogeneous Differential Equation ( $y = ux$ )**

Lecture 5 Slide 9

**Problem:** Solve  $(x^2 + y^2)dx + (x^2 - xy)dy = 0$ .**Step 1: Verify homogeneity** $M$  and  $N$  both have degree  $\alpha = 2$ .**Step 2: Substitution**Let  $y = ux \implies dy = u dx + x du$ .**Step 3: Substitute and simplify**

$$x^2(1 + u^2)dx + x^2(1 - u)(u dx + x du) = 0$$

Divide by  $x^2 \implies (1 + u)dx + x(1 - u)du = 0$ .**Step 4: Separate variables & integrate**

$$\frac{dx}{x} + \left(1 - \frac{2}{u+1}\right)du = 0 \implies \ln|x| + u - 2\ln|u+1| = C_1 \implies (u+1)^2 = Cxe^u$$

**Step 5: Back-substitute  $u = y/x$** 

$$(x + y)^2 = cx^3e^{y/x} \quad \text{and singular solution } y = -x$$

**★ Homogeneous Pattern to Remember:**

1. Verify  $M(tx, ty) = t^\alpha M$  and  $N(tx, ty) = t^\alpha N$ .
2. Substitute  $y = ux$  and  $dy = u dx + x du$  (product rule!).
3. Factor out  $x^\alpha$  and cancel.
4. Separate  $u$  and  $x$ , integrate, and back-substitute  $u = y/x$ .

**Problem 9 · Bernoulli's Differential Equation ( $u = y^{1-n}$ )**

Lecture 5 Slide 11

**Problem:** Solve  $x \frac{dy}{dx} + y = x^2 y^2$ .**Step 1: Put in canonical Bernoulli form**Divide by  $x$ :  $\frac{dy}{dx} + \frac{1}{x}y = xy^2$  ( $P(x) = 1/x$ ,  $f(x) = x$ ,  $n = 2$ ).**Step 2: Linearize using  $u = y^{1-n}$** With  $1 - n = 1 - 2 = -1$ , substitute  $u = y^{-1} = 1/y$ :

$$\frac{du}{dx} + (1 - n)P(x)u = (1 - n)f(x) \implies \frac{du}{dx} - \frac{1}{x}u = -x$$

**Step 3: Solve the linear ODE in  $u$** 

$$\mu(x) = \frac{1}{x} \implies \frac{d}{dx} \left[ \frac{1}{x}u \right] = -1 \implies \frac{1}{x}u = -x + C \implies u(x) = Cx - x^2$$

**Step 4: Back-substitute  $u = 1/y$** 

$$y(x) = \frac{1}{Cx - x^2} = \frac{1}{x(C - x)} \quad \text{and singular solution } y \equiv 0$$

**★ Bernoulli Pattern to Remember:**

1. Divide by leading coefficient to get  $\frac{dy}{dx} + P(x)y = f(x)y^n$ .
2. Use substitution  $u = y^{1-n}$  to instantly get  $\frac{du}{dx} + (1 - n)P(x)u = (1 - n)f(x)$ .
3. Solve the linear equation for  $u(x)$  via integrating factor.
4. Back-substitute  $y = u^{1/(1-n)}$ . Note  $y \equiv 0$  is a singular solution.



**Problem 10 · Reduction to Separation of Variables ( $u = Ax + By + C$ )**

Lecture 5 Slide 12

**Problem:** Solve  $\frac{dy}{dx} = (-2x + y)^2 - 7$ .**Step 1: Identify linear argument and substitute**

$$\text{Let } u = -2x + y \implies \frac{du}{dx} = -2 + \frac{dy}{dx} \implies \frac{dy}{dx} = \frac{du}{dx} + 2.$$

**Step 2: Transform to separable ODE**

$$\frac{du}{dx} + 2 = u^2 - 7 \implies \frac{du}{dx} = u^2 - 9$$

**Step 3: Check constant solutions**

$$u^2 - 9 = 0 \implies u = \pm 3 \implies y = 2x \pm 3$$

**Step 4: Separate variables & partial fractions**

$$\frac{du}{(u-3)(u+3)} = dx \implies \frac{1}{6} \ln \left| \frac{u-3}{u+3} \right| = x + C_1 \implies u(x) = \frac{3(1 + ce^{6x})}{1 - ce^{6x}}$$

**Step 5: Back-substitute  $u = -2x + y$** 

$$y(x) = 2x + \frac{3(1 + ce^{6x})}{1 - ce^{6x}} \quad \text{and singular solution } y = 2x - 3$$

**★ Linear Argument Pattern to Remember:**

1. Substitute  $u = Ax + By + C$ .
2. Differentiate:  $\frac{dy}{dx} = \frac{1}{B} \left( \frac{du}{dx} - A \right)$ .
3. Transform into separable ODE:  $\frac{du}{dx} = A + Bf(u)$ .
4. Separate, integrate, and back-substitute.